IC526 Advanced Compilers
Lec 02: OCaml Basics

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Why OCaml?

~~• Because the professor likes it.~~

~~• Because functional programming is superior.~~

- Functional programming with algebraic data types fit compiler development.
  - Language syntax, types, analysis, …
- Real compilers & code analyzers use it.
  - CompCert, Infer, Rust (early)
- Learning a new language is part of the point.
  - You'll see what a high-level language features can do for you.

Why OCaml?

- Algebraic data types + pattern matching = natural fit for compilers.

[expr]

- 3
- 4
- 12
- 3 + 12
- (3 + 12) + 4
- …

```
type expr = Num of int | Add of expr * expr


let rec eval e =
  match e with
  | Num n -> n
  | Add (e1, e2) -> eval e1 + eval e2
```

The core pattern you'll use throughout this course

Setup

- Follow the setup instructions on the course website.
  - Package setup – helps/Setup.md
  - IDE setup – helps/Editor.md
  - AI tools will help you troubleshoot.

- Bring your laptop for in-class exercises.

Run OCaml

- Interactive mode

  $ utop

- To run a single file (for practice)

  $ ocaml file.ml # uses interpreter

  $ ocamlc file.ml –o file && ./file # uses compiler

- **To build a project** – use *Dune*

  $ dune init proj my_proj # generates new project named “my_proj”

  $ dune exec bin/main.exe # build and run; for build-only: dune build

Hello, World!

$ dune init proj my_proj # create a test project ‘my_proj’

$ code . # if you use VS Code
# don’t miss the dot [.]

- Open bin/main.ml in VSCode

```
let () = print_endline "Hello, World!"
```

- Terminal → New Terminal
  $ dune exec bin/main.exe

Basic Features

Global *Let* Bindings

- An .ml file contains a sequence of **bindings**

```
let my_val = 2 + 3

let add n m = if n = 0 then m else n + m

let () = Printf.printf "Add my_val and 3: %d\n" (add my_val 3)
```

1. **RHS expression evaluates to a value**
2. **The LHS variable *is bound to* the value**

`Printf.printf` : A builtin *effectful* function that returns ***unit*** [()]

Forget about **memory & commands**!
Only ***expressions*** and ***global bindings.***

Local *Let* Bindings and *Scopes*

- [`let x = e1 in e2`] locally binds [x] to the value of [e1] in [e2].
- [x] is only visible in [e2]: "[e2] is the scope of [x]".

```
let x = 10                                  (* global binding *)

let rec fib n =                             (* recursive function needs [rec] *)
  if n == 0 then 0                          (* 'if' expression *)
  else if n == 1 then 1                     (* just a nested if *)
  else
    let x = fib (n - 2) in                  (* local binding – shadows global [x] *)
    let y = fib (n - 1) in                  (* local binding (nested) *)
    x + y

let () = Printf.printf "fib x = %d\n" (fib x)      (* fib(10) = 55 *)
```

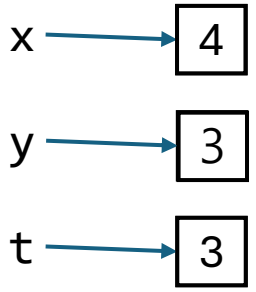
Binding ≠ Storing in Memory

C++ (Storing in memory)

- A variable is mapped to a cell's addr

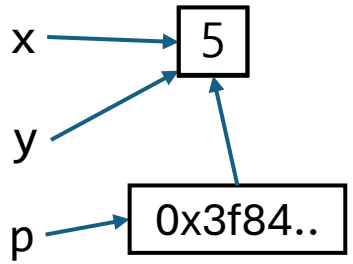
Update is allowed

```
// x = 3, y = 4
t = x; x = y; y = t;
// x = 4, y = 3
```



Multiple variables may point the same cell

```
// x = 3
int &y = x;
int *p = &x;
y = 4; // x = 4
*p = 5; // x = 5
```

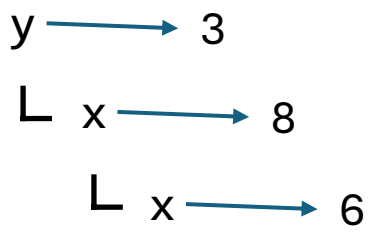


OCaml (Binding)

- A variable is mapped to a value

No updates – `let` only generates a new scope

```
let y = 3 in
let x = y + 5 in
(let x = y * 2 in
 x + 2) * x
```



"shadowing"

Why this design?

Not true in C++, in general (why?)

```
(fib x) + ..
.. (* no let x *)
(fib x) * ..
```

➡

```
let t1 = fib x in
t1 + ..
..
t1 * ..
```

Note: The concepts of shadowing & scope exist in C++, too; but their usage are more critical in Ocaml.

Scope Practice: Nested Let Bindings

```
let x = 5
```

```
let () =
    let y =
        let x = x + 1 in
        (let x = x * 3 in
         x + 2) + x
    in
    Printf.printf "%d\n" (x + y)
```

(Answer: 31)

Loops? Use Recursion!

C++

- Implement “is_prime”

```
bool is_prime(int n) {

  for (int i=2; i < n; ++i)
    if (n % i == 0)
      return false;

  return true;
}
```

OCaml

```
let rec is_prime_i i n =
  if i < n then
    if (n mod i == 0) then false
    else is_prime_i (i+1) n
  else true

let is_prime n =
  is_prime_i 2 n
```

It’s a Typed Language

- Every expression has a *type*, determined at **compile time**.
  - *Type errors* are caught **before** running the program.
- Types can be **inferred** — in many cases, you may omit them.
  - Still, type annotations help **improve readability** and **reduce errors**.

```
let x : int = 10          (* x : int *)
let x2 = 10               (* x2 : A => Since 10 : int, A = int *)


let double (n : int) : int = (* double : int -> int *)
  let y : int = n * 2 in
  y

let double2 n =           (* double2 : A -> B, and n : A *)
  let y = n * 2 in        (* 2 is int => A = int, therefore, y : int *)
  y                       (* y : int => B = int *)
```

It’s a Functional Language

- Functions are **first-class values** — they can be passed around like any other value.

```
let apply_twice (f: int -> int) (x: int) : int =   (* function parameter *)
  f (f x)

let add (n m: int) : int = n + m

let () =
  let () = Printf.printf "double (double 2) = %d\n" (apply_twice double 2) in
  let add3 k = add 3 k in                          (* local function *)
  let () = Printf.printf "add3 (add3 5) = %d\n" (apply_twice add3 5) in
  let () = Printf.printf "add2 (add2 5) = %d\n" (apply_twice (add 2) 5) in
                                                   (* partial application *)
  Printf.printf "add2 (add2 5) = %d\n" (apply_twice (fun x -> x + 2) 5)
                                                   (* anonymous function *)
```

[;] : Syntactic Sugar

- [e1; e2] is a shorthand for [let () = e1 in e2]

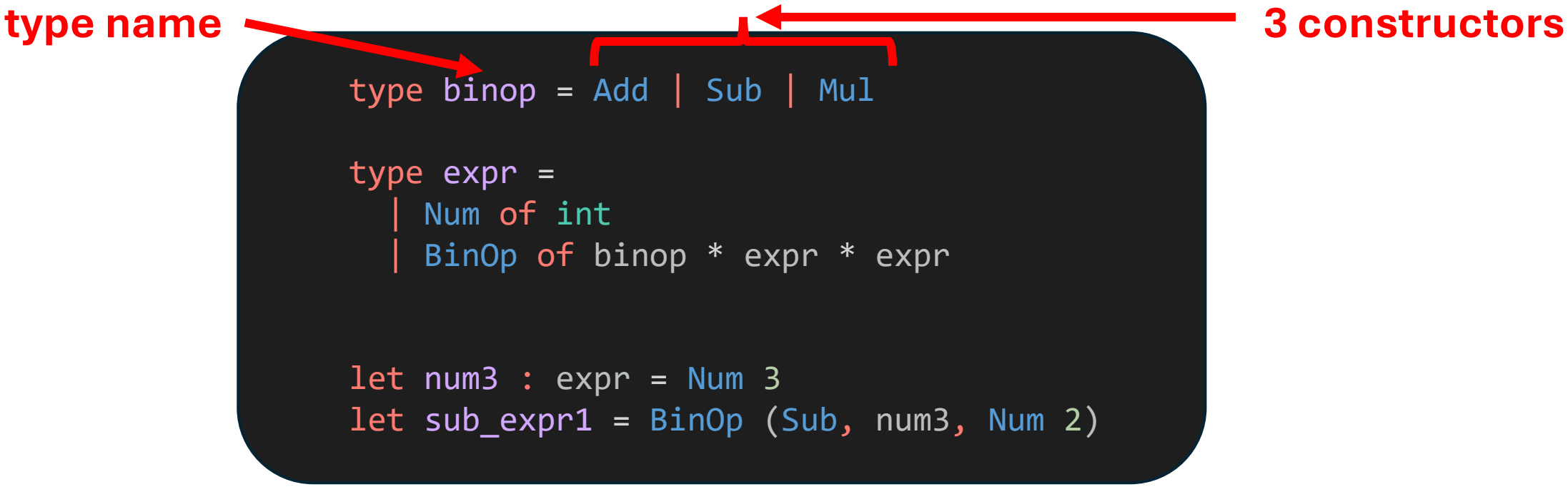
```
let apply_twice (f: int -> int) (x: int) : int =   (* function parameter *)
  f (f x)

let add : int -> int -> int = fun n m -> n + m     (* the same definition *)

let () =
  let add3 k = add 3 k in
  Printf.printf "double (double 2) = %d\n" (apply_twice double 2);
  Printf.printf "add3 (add3 5) = %d\n" (apply_twice add3 5);
  Printf.printf "add2 (add2 5) = %d\n" (apply_twice (add 2) 5);
  Printf.printf "add2 (add2 5) = %d\n" (apply_twice (fun x -> x + 2) 5)
```

ADT and Pattern Matching

Algebraic Data Types (ADT)



- Each case has a *constructor* of the type.
- Constructors may carry data.

Pattern Matching – How to “Use” ADTs

```
type binop = Add | Sub | Mul

type expr =
  | Num of int
  | BinOp of binop * expr * expr


let rec eval e =
  match e with
  | Num n -> n
  | BinOp (op, e1, e2) ->
    match op with
    | Add -> eval e1 + eval e2
    | Sub -> eval e1 - eval e2
    | Mul -> eval e1 * eval e2
```

- Exhaustiveness check — **compiler warns** if you miss a case.

Pattern Matching – How to “Use” ADTs

```
type binop = Add | Sub | Mul

type expr =
  | Num of int
  | BinOp of binop * expr * expr


let rec eval' e =
  match e with
  | Num n -> n
  | BinOp (Add, e1, e2) -> eval' e1 + eval' e2
  | BinOp (Sub, e1, e2) -> eval' e1 - eval' e2
  | BinOp (Mul, e1, e2) -> eval' e1 * eval' e2
```

- Patterns can be nested.

List: A Widely-Used ADT for Sequences

```
let xs : int list = [1; 2; 3]
let es : expr list = [Num 3; BinOp (Mul, Num 1, Num 2)]

let ys = 0 :: xs          (* [0; 1; 2; 3] *)
let zs = xs @ [4; 5]      (* [1; 2; 3; 4; 5] *)

let ys = List.map double xs (* [2; 4; 6] *)
```

```
type 'a list =
| []
| (::) of 'a * 'a list

let rec map (f: 'a -> 'b) (l: 'a list) : 'b list =
  match l with
  | [] -> []
  | a :: l' -> (f a) :: (map f l')
```

Option Type: Representing Failures

```
type 'a option =
| None
| Some of 'a


let rec nth (l: 'a list) (n: int) : 'a option =
  match l with
  | [] -> None
  | h :: t -> if n = 0 then Some h else nth t (n-1)
```

Ready To Go!

Exercises: HW0

- Create an ml project using Dune.
- Open hw_files/hw0.ml in the course website.
- Copy and past the content of hw0.ml into your main.ml
- Try now!

Linking multiple .ml Files (by Dune)

def.ml

```
type binop = Add | Sub | Mul

type expr =
  | Num of int
  | BinOp of binop * expr * expr


let rec eval e =
  match e with
  | Num n -> n
  | BinOp (op, e1, e2) ->
    match op with
    | Add -> eval e1 + eval e2
    | Sub -> eval e1 - eval e2
    | Mul -> eval e1 * eval e2
```

main.ml

```
let expr : Def.expr = Def.Num 3

let result = Def.eval expr
```

Or,

main.ml

```
open Def

let expr : expr = Num 3

let result = eval expr
```

More to Learn!

- Exceptions
- Records
- References (not necessary for this course)

Learn By Exercise: Assignment #1

- Build a tiny compiler: **arithmetic expressions → stack machine.**

Source language

```
1 + 2 * 3
```

Target: a stack machine

```
PUSH 1;  PUSH 2;  PUSH 3;  MUL;  ADD
```

(1) Write a compiler

(2) Write a “simulator” – an interpreter of the stack language

(3) Write an interpreter using the option type

(4) Write an interpreter using exceptions

How To Do Assignments

- Check the notice – leave your ID and name until next Monday.
  - You will be given a private GitHub repository.
  - The repository contains the skeleton project.

- For each assignment, the specification document will be posted on the course website.
  - Check the course website.
  - Push your homework in time.